



Thorben Foerst

DevOps & Cloud Engineer

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Summary

IT Management student (B.Sc.) at the Technical University of Cologne, specialising in cloud and DevOps topics. AWS Certified Solutions Architect – Associate, with practical experience in setting up Kubernetes clusters using Terraform and in working with GitOps workflows via ArgoCD.

Skills

Cloud (AWS)

VPC, IAM, EC2, S3, RDS, EKS, Secrets Manager and many more (SAA-C03 certified)

Containers & Orchestration

Kubernetes, k3s, Amazon EKS, Docker, Docker Compose, Helm

IaC & CI/CD

Terraform, GitHub Actions, ArgoCD (GitOps), Ansible, Git

Monitoring & Observability

Prometheus, Grafana, Uptime Kuma

Network & Infrastructure

VLANs & subnetting, DHCP/DNS/firewall configuration, reverse proxies (Traefik, Nginx Proxy Manager), load balancing, WireGuard

Programming

Python, HCL, Bash, YAML, JSON, SQL, TeX, C#, C++

Miscellaneous

Linux, RAG, LangChain, Chroma, Proxmox, Hetzner, TrueNAS, GitHub, LaTeX, MikroTik, Longhorn, NetBird VPN

Certifications

AWS Certified Solutions Architect – Associate

SAA-C03 – May 2026

Credly verification

Languages

German (native language)

English (B2)

Education

B.Sc. in IT Management

TH Köln – Campus Gummersbach

Oct. 2022 – expected Feb. 2027

- Bachelor's thesis: *"GitOps vs. Traditional CI/CD in a Kubernetes Context: A Comparative Evaluation Study Using ArgoCD and GitHub Actions"*

Experience

Software Developer (Work Placement Semester)

Foerst GmbH

Oct. 2025 – March 2026

- Design and development of interactive training modules for driving simulation software
- Development of an in-house RAG-based chatbot (Streamlit, OpenAI API, ChromaDB) that supports developers in creating new exercise modules by providing context-specific knowledge
- Containerisation and deployment of applications on on-premises servers using Docker and Docker Compose
- Induction and professional support for new staff

Software developer

Foerst GmbH

Oct. 2021 – Sep. 2022

- Development of interactive training modules for driving simulation software
- Support with the delivery of driving simulators
- Production support

Projects

Practical project work

TH Köln, module in the final semester, forming the basis for the Bachelor's thesis

Apr. 2026 – Sep. 2026

- Setting up a production-ready Kubernetes infrastructure in the AWS Cloud using Terraform, including the deployment of a managed Kubernetes cluster (Amazon EKS)
- Containerisation and deployment of a test application with a database connection (Amazon RDS) for end-to-end validation of the infrastructure
- Secure management of database access credentials using AWS Secrets Manager, and provisioning via AWS Load Balancer Controller (LBC)

Home Lab

Personal project

June 2025 – ongoing

- Setting up a custom-designed network with its own routing and VLAN segmentation as the basis for the entire infrastructure
- Virtualisation layer on two Proxmox nodes; on top of this, a Kubernetes cluster (k3s) with multiple master and worker nodes to ensure resilience against individual node failures
- GitOps deployment via ArgoCD, Longhorn storage with automated S3 backups for services such as Nextcloud, Immich and Vaultwarden
- Infrastructure-as-Code using Terraform, and monitoring with Prometheus and Grafana